

Incremental Graph Layout by Stochastic Search: a Routed Objective, a Calibrated Metric, and a Matched-Budget Control

Vasili Gavrilov*

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Abstract

Placing the nodes of a graph in the plane with few edge crossings is NP-complete, so every layout tool runs heuristics, and the heuristics are rarely explicit about three things: what objective they optimize, whether that objective measures the drawing the reader actually sees, and whether the search spends its budget any better than uniform random sampling would. We treat all three. The problem is formulated incrementally: a graph grows by k nodes and the existing drawing is frozen, which keeps the reader’s mental map, bounds the search space at $2k$ variables, and contains the classic full redraw as the limit of an anchor term that prices the displacement of frozen nodes. The objective reads the drawing’s real medium, orthogonally routed connectors rather than the straight center-to-center segments most objectives score, and the metric is calibrated: the router’s shortfall against a reference layout of certified on-screen quality is measured and printed beside every number that depends on it. Four families of stochastic search (hill climbing with restarts, simulated annealing, a genetic algorithm, and uniform random sampling as the matched-budget control) are compared by exact sign tests on paired seed panels, deterministically reproducible to the byte across platforms. The motivating application is the entity–relationship diagram, which mainstream tools still place badly enough that maintainers redraw by hand; a proof of concept places the tables added by [PENDING: n] real consecutive schema migrations, each with a human-accepted answer, under a pre-registered analysis. [PENDING: primary result sentence]. One general finding: under any objective of this family the searches outscore the human reference, which measures the objective’s blindness to what maintainers optimize rather than any superiority of the tool, and we argue this gap is why automatic graph layout has stayed bad in practice.

1 Introduction

[PENDING: introduction]

2 The problem

[PENDING: formulation]

3 The objective is the medium

[PENDING: objective]

*Independent researcher. Code, data and derived tables: <https://github.com/Anode1/cjitter>.

4 The searches and the control

[PENDING: library]

5 The benchmark

[PENDING: benchmark]

6 Pre-registered experiment

[PENDING: results]

7 What the human comparison measures

[PENDING: discussion]

8 Related work

[PENDING: related]

9 Conclusion

[PENDING: conclusion]